

NIPPUN SABHARWAL

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EDUCATION

University of Illinois Urbana-Champaign

Aug 2022 – May 2026

Bachelor of Science in Computer Engineering

Selected coursework: Deep Learning for Computer Vision (A), Applied Machine Learning (A), LLM Engineering Design (A), Robotic Manipulation (A), Digital Systems Laboratory (A)

RESEARCH EXPERIENCE

Research Assistant, Vision and Robotics Laboratory / NCSA

Jan 2025 – Present

Prof. Narendra Ahuja, Research Professor, CSL, UIUC

- Independently developed an unsupervised, physics informed pipeline that reconstructs planar scenes from video sequences captured through turbulent refractive media.
- Implemented a 3D water surface reconstruction pipeline, with a vision transformer based encoder and NeRF style positional encodings to jointly model imaging physics and correct non rigid video distortions. Enforced physical constraints with a differentiable wave based simulator and with spatial and temporal power spectrum losses.
- Built an image recovery pipeline driven by surface slope estimation from input video that stitches correct pixels to within decibels of ground truth baseline. The estimator is based on learned trajectory statistics from millions of voxels, using interpretable features such as multi-spectral edges, moment of inertia and symmetry mapped to low slope probability.

Research Assistant, 3D Reconstruction and Policy Evaluation

Aug 2025 – Present

Prof. Svetlana Lazebnik, Siebel School of Computing, UIUC

- Independently reconstructed the ALOHA teleoperation setup using 3D Gaussian splatting to obtain a photorealistic three dimensional representation for visualizing and debugging robotic policies in real to simulation experiments.
- Integrated splatted scenes with manipulation policy rollouts, building scripts to render multiview videos and visually compare policy variants, failure modes, and generalization across camera viewpoints.

Research Assistant, VR Teleoperation and Imitation Learning

Aug 2024 – Jan 2025

Prof. Joohyung Kim, Department of ECE, UIUC (ECE 398)

Demo: <https://nippun-live.github.io/>

- Built Walle, a virtual reality teleoperation interface using Meta Quest 2 head tracking and inverse kinematics, enabling low latency internet teleoperation for long horizon tasks such as sorting, insertion, and folding.
- Implemented 3 degree of freedom head tracking and an inverse kinematics neck controller that streams 60 Hz joint commands and RGBD video with about 40 to 60 millisecond round trip latency and sub degree motion accuracy.
- Ran small user studies showing roughly 30 percent faster fine manipulation with Walle, and maintained the codebase, logging infrastructure, and demonstration dataset for downstream imitation learning.

Research Assistant, Multimodal LLMs for Engineering Design

Jan 2025 – Jun 2025

Prof. Bin Hu (ECE 498)

- Co led the design of an open source benchmark for evaluating multimodal large language models on engineering design tasks spanning more than 500 test cases with simulation based grading of physical feasibility and performance. The tasks require joint understanding of diagrams, timing waveforms, and textual specifications, to evaluate performance on spatially grounded engineering problems such as constraint satisfaction and temporal behavior.
- Led the Digital System Design track, building automated HDL simulation and scoring pipelines with SystemVerilog, Icarus Verilog. Ran large scale experiments on commercial and open models and contributed rubric design, task construction, and failure mode analysis, accepted to **NeurIPS 2025**.

PUBLICATIONS

- X. Guo, Y. Li, N. Sabharwal, S. Sinha, et al., “Toward Engineering AGI, Benchmarking the Engineering Design Capabilities of LLMs.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- N. Sabharwal, N. Ahuja, “Unsupervised Recovery of Planar Surfaces Under Turbulent Media Using Geometric Reconstruction.” *Senior Undergraduate Thesis, Department of ECE, University of Illinois Urbana Champaign*
- N. Sabharwal and N. Ahuja, “Unsupervised Removal of Water Turbulence”, *manuscript in preparation for submission to ECCV 2026*.
- J. Allen and N. Sabharwal, “Using Reciprocals of Prime Numbers to Label Proteins,” *submitted to Bioinformatics (Oxford University Press)*

INDUSTRY EXPERIENCE

Intern, World Models and Hardware

Jul 2025 – Aug 2025

Stealth Robotics Startup, San Francisco Bay Area

- Implemented the **flow matching** head for a latent autoregressive world model that learns visual dynamics from unlabeled robot video and supports action conditioned rollouts.
- Led performance tuning of the world model training stack, developing mixed precision CUDA kernels and profiling workflows that increased H100 training throughput by about 2.3 times and significantly reduced GPU memory usage, stabilizing long horizon training runs.

AI and Data Science Intern, Global Supply Chain

May 2025 – Aug 2025

AstraZeneca, Gaithersburg, MD

- Developed an intelligent agent with a natural language interface and retrieval over supply, demand, inventory, and logistics records, generating automated insights and what if scenarios that informed allocation and inventory decisions.
- Architected a machine learning production system end to end, implementing an MLOps pipeline and GPU accelerated services in CUDA that serve more than ten million supply chain data points with under 200 millisecond latency.

TECHNICAL SKILLS

Programming Languages Python, C, C++, CUDA, Verilog, SystemVerilog, RISC V Assembly, Java, SQL

3D Vision and Graphics NeRF, 3D Gaussian Splatting, COLMAP, PyTorch3D, Open3D, OpenGL, OpenCV

Robotics and Generative Models Transformers, Diffusion and Flow Matching Models, BERT, VAEs, TensorRT, vLLM, mixed precision, LoRA, RAG

HONORS

- Compall Family Scholarship (2025 to 2026)
- Samsung Engineering Scholarship (2024 to 2025)
- Illinois Engineering Achievement Scholarship (2023 to 2024, 2025 to 2026)
- Illinois Engineering Outstanding Scholarship (2023 to 2024, 2025 to 2026)
- James Scholar Honors (2022 to 2023)

TEACHING

Undergraduate Teaching Assistant, ECE 110 Introduction to Electronics

Aug 2023 – May 2025

- Graded weekly homeworks and led lab sections on measurement, modeling, and analysis of circuits involving diodes, CMOS logic, BRAM, amplifiers, and other components for first year students.

Undergraduate Teaching Assistant, ECE 120 Introduction to Computing

Jan 2024 – Dec 2024

- Grading weekly homeworks for digital logic, finite state machines, and introductory computing projects, helping students debug designs and understand timing and state behavior.

SELECTED PROJECTS

16-bit RISC CPU and Graphics Controller (<https://github.com/nippun-live/svcpu>)

Spring 2024

- Designed and verified a 16 bit RISC CPU on a Xilinx Spartan 7 FPGA with a three stage pipeline, ALU, register file, and control logic, meeting a 50 MHz timing target with about 3 nanoseconds of slack through self checking testbenches.
- Created a memory mapped input output subsystem over on chip BRAM linking UART, switches, and seven segment displays, and added text based color graphics and simple screensavers by designing an AXI4 HDMI text controller that runs at 60 frames per second.

RISC V Operating System Kernel and IO Drivers (<https://github.com/nippun-live/riscvos>)

Fall 2024

- Built a UNIX style kernel in C with bootloader, trap handling, Sv39 demand paged virtual memory, a preemptive scheduler, and more than 20 user mode system calls, sustaining stable multitasking workloads in QEMU.
- Designed a write ahead logged, block based file system with a configurable write back cache that preserves crash safe metadata consistency across VirtIO and in memory devices, and implemented drivers for UART, RTC, PLIC, VirtIO, GPIO, and SPI along with an ELF loader that can run interactive applications and games.